

Space Shooter - Setup and Build Guide (Unity 2022.3 LTS)

This guide is the full step-by-step process to open, configure, and build the project as a Windows executable.

1) Set up the Unity project

1. Install **Unity Hub**.
 2. Install **Unity 2022.3 LTS** editor (2D module + Windows Build Support/IL2CPP).
 3. In Unity Hub, click **Open** and select this project folder.
 4. Wait for import and script compilation.
 5. Confirm project uses **Built-in Render Pipeline**.
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2) Import scripts and assets

All code and placeholder assets are already in the repo:

- Scripts: `Assets/Scripts/`
- Sprites: `Assets/Sprites/`
- Audio: `Assets/Audio/`

Required Tags

Go to **Edit > Project Settings > Tags and Layers** and ensure these tags exist:

- `Player`
- `Enemy`
- `PlayerBullet`
- `EnemyBullet`
- `PowerUp`

Input

This project uses Unity classic input axes:

- `Horizontal`
- `Vertical`

And key polling:

- `Space` (shoot)
 - `Escape` (pause)
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3) Configure scenes and prefabs

Scene configuration

Use:

- `Assets/Scenes/SCENE_CONFIGURATION.md`

Create two scenes in Unity:

- MainMenuScene.unity (optional split menu scene)
- GameScene.unity (main gameplay scene)

Add at minimum in gameplay scene:

- Main Camera (+ ScreenBounds script)
- GameManager object (+ GameManager script)
- AudioManager object (+ AudioManager script + audio clips)
- SpawnManager object (+ SpawnManager script + enemy prefab refs)
- Player object (+ PlayerController script)
- UICanvas + UIManager object (+ UIManager script)
- StarField and ParallaxBackground

Prefab configuration

Use:

- Assets/Prefabs/PREFAB_SETUP.md

Create and assign these prefabs:

- PlayerBullet
- EnemyBullet
- EnemyBasic
- EnemyZigzag
- EnemyTank
- PowerUpHealth
- PowerUpRapidFire
- PowerUpShield

Wire references exactly as documented in prefab setup.

4) Build game as Windows .exe

1. Open **File > Build Settings**.
2. Select **PC, Mac & Linux Standalone**.
3. Target Platform: **Windows**.
4. Architecture: **x86_64**.
5. Add gameplay scene(s) to **Scenes In Build**.
6. Open **Player Settings** and set:
 - Product Name: Space Shooter
 - Default resolution: 1024 x 768 (or preferred)
 - Fullscreen mode: Windowed
7. Click **Build**.
8. Choose output folder (for example Builds/Windows).
9. Run generated Space Shooter.exe .

Sanity checklist before building

- No compile errors in Console.

- All script references assigned in Inspector.
- All required tags exist.
- SpawnManager has all enemy prefab references.
- Player has bullet prefab + fire point.
- Enemy prefabs have enemy bullet prefab + fire point.
- UIManager panel and text/button references are connected.
- Audio clips assigned in AudioManager.

Note

If you copy this project to another machine, open with Unity 2022.3 LTS and let Unity reimport all assets before building.